

| Defense (per backbone) | AD-std ASR (%)↓   | AD-benign util. (%)↑ | Setup                          |
|------------------------|-------------------|----------------------|--------------------------------|
| <i>Llama-3.1-8B</i>    |                   |                      |                                |
| No defense             | 8.7               | 20.3                 | 1×                             |
| <b>ODILE</b>           | <b>0.2</b> (−8.5) | 20.2 (−0.1)          | LoRA, 1×                       |
| <i>Qwen-2.5-7B</i>     |                   |                      |                                |
| No defense             | 9.6               | 35.5                 | 1×                             |
| <b>ODILE</b>           | <b>0.1</b> (−9.5) | 35.6 (+0.1)          | LoRA, 1×                       |
| <i>Qwen-2.5-14B</i>    |                   |                      |                                |
| No defense             | 8.3               | 46.1                 | 1×                             |
| CaMeL <sup>‡</sup>     | 0.0 <sup>‡</sup>  | 0.0 <sup>‡</sup>     | restricted-Py interpreter, ~6× |
| <b>ODILE</b>           | <b>0.0</b> (−8.3) | 41.8 (−4.3)          | LoRA, 1×                       |
| <i>Qwen3-8B</i>        |                   |                      |                                |
| No defense             | _ <sup>†</sup>    | 21.75                | 1×                             |
| <b>ODILE</b>           | _ <sup>†</sup>    | 18.25 (−3.5)         | LoRA, 1×                       |
| <i>Qwen3-32B</i>       |                   |                      |                                |
| No defense             | 8.1               | 35.3                 | 1×                             |
| <b>ODILE</b>           | <b>0.5</b> (−7.6) | 37.5 (+2.2)          | LoRA, 1×                       |

Table 2: **Cross-model AgentDojo: five backbones × {base, ODILE}**. ASR (% , ↓) is averaged across the four AgentDojo suites under AD-standard attack injections ( $n=8,177$  pairs per cell). Benign utility (% , ↑) is the agentic benign-no-attack rate (suite-averaged,  $n=629$  per cell). Bracketed numbers next to ODILE are absolute deltas in percentage points vs. that backbone’s no-defense row. ODILE drives ASR to  $\leq 0.5\%$  on every backbone measured and preserves benign utility within  $\pm 4.3$  pp of base across the five smaller backbones (median delta  $\approx 0$  pp). The Llama-3.3-70B headline with the full comparator set is Table ??; the visual cross-model summary is Figure ?? Panel A. Per-suite and per-attack-family breakdowns in Appendix ??. <sup>†</sup>Qwen3-8B AD-benign is reported here; the AD-standard attack sweep on Qwen3-8B is not yet complete — the wiring-validation smoke (banking, **direct** attack,  $n=144$ ) reports 0% ASR for both base and ODILE on Qwen3-8B (Appendix ??), but the full 52-cell grid is pending. <sup>‡</sup>CaMeL [ref] is a design-level defense that splits the agent into a privileged-planner LLM and a quarantined LLM communicating through a restricted-Python interpreter. At Qwen-2.5-14B as the privileged LLM, the model cannot drive the interpreter loop (0/5 banking benign tasks; 0/18 attacked cells); the privileged LLM either refuses outright (“I don’t have the capability”), hallucinates output strings, or loops in a “please provide the code” debug regime. CaMeL’s 0% ASR at this scale is structurally degenerate (no tool calls — benign or attacker-directed — are produced) and cannot be credited to the defense mechanism. The original CaMeL evaluation uses frontier-grade closed-API models (Gemini-Pro / Claude-3.5 / GPT-4); ODILE achieves the same 0% ASR at this open-weights scale while preserving 41.8% benign utility (Appendix ??).